# Design and Cost Modeling Document

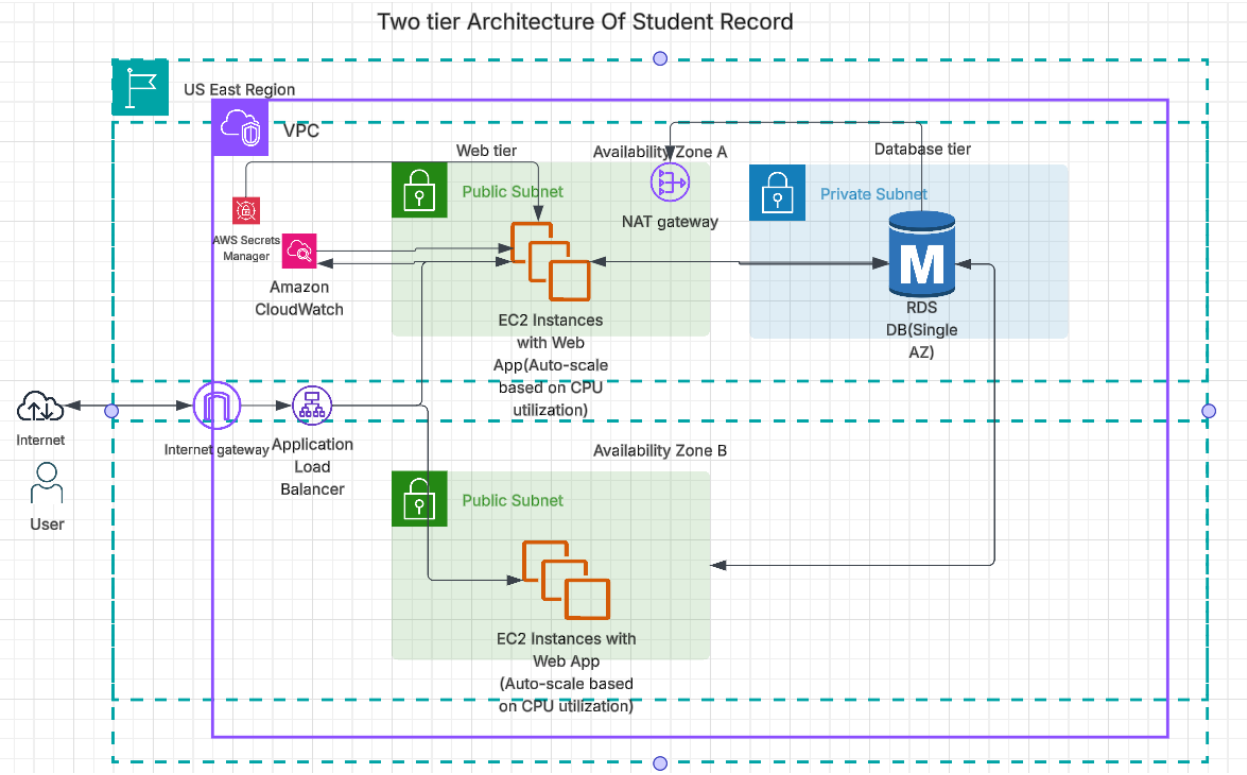
Max. Points: 200

The purpose of this document is to guide students in designing a complete cloud-based solution in AWS while developing an understanding of cost awareness and financial decision-making in cloud environments. This assignment requires students to create an architectural diagram that represents a well-structured cloud solution, identify the necessary cloud services and specifications, and estimate associated costs using the AWS Pricing Calculator tool.

Be sure to check back with the given problem statement and your Requirements Document to ensure any important detail is not missing.

### 1: Architectural Diagram (50 points)

Paste your Architectural Diagram in the space below. Each of the components/services should be correctly labeled, and the traffic flow patterns should be depicted.



## 2: Cost Calculations and Justification (50 points)

Given Information/Assumptions: **(5 points)**

**Note: Use values provided in the problem statement or discussed during the Requirements meeting. Any missing values may be reasonably assumed based on the given criteria.**

1. Average visitors/day to the website = 2000
2. No. of hours the website is active per day = 12
3. Average visit time in minutes per user = 10
4. Average page views per visitor = 3
5. Average page size for the website in MB = 2

Use *Formulas for Cost Calculations on AWS* document as the reference to determine the following parameters.

1. Determine the number of Concurrent users (CCU) for the website. Calculations must be shown at each step to receive credit. **(10 points)**
   2000 visitors/day / 12h = 167 visitors/h / (60min / 10min visits) = **28 CCU**

2. Specify the choice of Amazon EC2 instance type / size with justification. **(5 points)**
   **t3.small** - For an average CCU of 28, one t3.medium instance could handle the traffic, but the customer has specified a minimum of two instances be deployed at all times to support the high availability requirement. Two t3.small instances can handle the average load of 28 CCU as one instance is recommended for a CCU between 10 and 20 and a load balancer will distribute the load between the two.

3. What is the minimum number of web instances you plan to deploy to support typical load? Explain why. **(7.5 points)**
   **Two instances** will be deployed at a minimum in different availability zones to support the high availability requirement. Two t3.small instances should be capable of serving the web application to our average 28 CCUs without needing to scale up.

4. What is the max. number of instances that should be deployed? Explain why. **(7.5 points)**
   **Fourteen instances** should be allowed maximum due to customer's expected maximum of 10,000 to 20,000 users per day. Less than ten t3.small instances should cover the lower end of this range with ease while fourteen should be enough

to handle the absolute maximum expected traffic during the peak season, based on a second CCU calculation using the peak daily user numbers and the maximum CCU for the recommendation of a t3.small instance.

5. Determine the bandwidth for the website. Calculations must be shown at each step to receive credit. **(15 points)**
   2000 visitors/day * 30 = 60,000 visitors/mo * 3 pages/visit = 180,000 pages/mo * 2MB / page = 360,000 MB/mo / 1000 = 360GB/mo * 1.5 for overhead = **540GB/mo**

### 3: Component Summary (50 points)

Fill out the table below to provide summarized information about the essential architectural components / services that will be used in the project. You will use information derived from your calculations as well as from the **Other Design Considerations for Costing** section of the *Formulas for Cost Calculations on AWS* document.

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| **AWS Component / Service** | **Specification** | **Description** |
|---|---|---|
| Amazon EC2 (web server) | Operating system: Linux<br>Instance type/size: t3.small | 100% utilization, On-demand pricing<br>On average per day, there are 2 instances running<br>Max. capacity: 14 |
| | One EBS volume per EC2 instance | Type: gp3<br>Amount: 8 GB |
| Data Transfer | Data IN from Internet: 1 TB/month<br>Data OUT to Internet: 540GB/mo | |
| Elastic Load Balancer | Type: Application Load Balancer (HTTP) | 1 GB/hour processed bytes (EC2 instances as targets) |

| | | |
|---|---|---|
| Amazon RDS (database server) | DB Engine: MySQL<br><br>Instance type/size: db.t3.micro (include vCPUs and memory) | Type of Deployment: Single-AZ<br><br>100% utilization, on-demand<br><br>RDS Proxy enabled |
| | EBS storage for RDS | Type: gp2<br><br>Amount: 20GB |
| Specify the AWS service you will deploy to ensure that the DB passwords are not stored on the web tier. | AWS Secrets Manager | Number of secrets: 1<br>Average duration of each secret: 30 days<br>API calls: 1 per month |

Table 1: AWS Component / Service Summary

### 4: Cost Estimate (50 points)

Use the AWS Pricing Calculator to calculate the monthly cost estimate for the essential AWS components / services in your architecture.

For costing purposes, work out EC2 costs for typical load only.

When working with the AWS Pricing Calculator, enter your team number or name in the description field. Export your estimate as a PDF (.CSV format is not acceptable) and upload separately to submit. You DO NOT need to work out costs related to database export or Azure services.